Haotong Luan

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Education

Shenzhen Technology University (SZTU) Sept 2024 – Present
B.Eng. in Automation, Future Technology School; GPA: 84.17/100

Research focus: underactuated grippers; environment-adaptive robotic hands; mechanism design.

Tsinghua University – Shenzhen X-Institute Sept 2025 – Present
P-SRT Program, Joint Training (Underactuated Robotics)

Advisor: Associate Prof. Wenzeng Zhang. Track: X-idea → **P-SRT (current)** → E-SRT → ORIC → SURF.

Publications and Accepted Papers

Legend: * Corresponding author

Peer-Reviewed Publications and Accepted Papers

DPS Hand: A Dual-Parallelogram and Slot-Guided Gripper for Adaptive Grasping and Environmental Scooping
Haotong Luan, Haokai Ding, Tao Yang*, Wenzeng Zhang*
Accepted to IEEE CASE 2026, RAS major conference, Shenyang, China, August 17–21, 2026. 2026

PlaceRecover: Layout-Invariant Point Cloud Recovery for Robust 3D Perception under LiDAR Placement Shifts
Benwu Wang, Zihang Wang*, Wenkai Zhu, **Haotong Luan**, Dong Kong, Binghao Wang, Yiming Zhou, Kailin Lyu, Teng Yan, Haoyu Wang, Yaohua Liu, Qiang Zhang
Accepted to IEEE/RSJ IROS 2026, top robotics conference, Pittsburgh, PA, USA, September 27–October 1, 2026. 2026

CSD: Conformal Selective Diagnosis for Reliable Bearing Fault Identification Under Domain Shift
Xi Yu, Anran Lu, Keyi Chen, **Haotong Luan***, Yuan Gao*
Accepted to IEEE SMC 2026, CCF C, Bellevue, WA, USA, October 4–7, 2026. (Corresponding author) 2026

Domain-Semantic Subspace Stacking for Financial Fraud Detection
Haotong Luan, Minmin Chen, Zhenhui Lin, Haokai Ding, Yiwei Sheng*
Accepted to ICIC 2026, CCF C, Toronto, Canada, July 22–26, 2026. 2026

An Underactuated Dual-Slide Adaptive Robotic Hand with Strong Environmental Adaptability
Haotong Luan, Wenzeng Zhang*, Tao Yang*
Proc. RAAI 2025, Singapore, Dec. 18–20, 2025. DOI: 10.1109/RAAI67517.2025.11423404 2025

An Idle-Stroke Underactuated Gripper with Independent Double-Parallelogram Linkages for Pinching and Adaptive Grasping
Yizhe Chen, **Haotong Luan**, Haokai Ding, Wenzeng Zhang*
Proc. RAAI 2025, Singapore, Dec. 18–20, 2025. DOI: 10.1109/RAAI67517.2025.11423397 2025

GLINT: An Idle-Stroke Grasp-and-Lift Hand for In-Hand Manipulation
Haokai Ding, **Haotong Luan**, Tao Yang*, Wenzeng Zhang*

Proc. ICCMA 2025, Paris, France, Nov. 24–26, 2025. DOI: 10.1109/ICCMA67641.2025.11369602 2025

Journal Manuscripts Submitted / Under Review

Manuscript submitted to *Advanced Intelligent Systems*
Haotong Luan, Jie Zhang, Yuan Gao
Submitted. Intelligent systems and robotics-related journal; JCR Q1. 2026

Manuscript under review at *Scientific Reports*
Haotong Luan, Xi Yu, Anran Lu, Keyi Chen, Jianwu Chen
Under review. JCR Q1. 2026

Patents Pending

- **3 invention patents** filed and under review.
- **1 utility model patent** filed and under review.

Research & Projects

Underactuated Robotic Gripper Research 2025 – Present
Shenzhen X-Institute, Tsinghua University
Designed underactuated grippers using linear-motion linkages, parallelogram mechanisms, and idle-stroke transmissions. Focus: embedding grasping logic into hardware for reliable single-actuator grasping. Led the DS Hand (first-author, RAAI 2025); collaborated on idle-stroke and GLINT gripper designs.

Visual-Servoing Control for Robotic-Arm Grasping 2026 – Present
Shanghai AI Laboratory (Remote RA)
Collaborating with Researcher Yuan Gao on visual-servoing control algorithms and end-effector design for robotic-arm grasping. Responsible for the mechanical gripper design mounted on the arm.

Quantum & AI Winter School Jan 2026
Shenzhen Loop Area Institute
Studied generative-model-based quantum variational circuit simulation (MLP / LSTM / Transformer) under Prof. Peng Zhang (Tianjin University). Worked on state fidelity optimization, mixed-state simulation, and physics-constrained training.

Experience

P-SRT Researcher – Shenzhen X-Institute Sept 2025 – Present
Advisor: Associate Prof. Wenzeng Zhang

Independently proposed the dual-slide adaptive gripper concept; conducted mechanism analysis and feasibility study, leading to long-term advising.

Remote Research Assistant – Shanghai AI Laboratory Apr 2026 – Present
Supervisor: Researcher Yuan Gao

Developing visual-servoing control algorithms for robotic-arm grasping and designing the mechanical end-effector.

Honors & Awards

National-Level Project (Principal Investigator), 2026 College Student Innovation and Entrepreneurship Training Program — "Zero-Lash Smart Grasping" 2026

Provincial-Level Project (Member), 2026 College Student Innovation Training Program — "Multi-Core Fiber Dynamic Optical Trap for Single-Cell Pathology" 2026

Crystal Prize for Scientific & Technological Innovation (top 10 across SZTU) 2025

Dean's Special Award for Rising Talent, Zhenxiong (Future Technology School, SZTU) 2024–2025

Outstanding Student Cadre, Shenzhen Technology University 2024, 2025
Excellence Prize, 15th "Challenge Cup" Student Entrepreneurship Competition (university round) .. 2025

Service & Leadership

- **Principal Investigator**, National-Level Project "Zero-Lash Smart Grasping", 2026 College Student Innovation and Entrepreneurship Training Program (2026.05 – present).
- **Project Member**, Provincial-Level Project on multi-core fiber dynamic optical trap for single-cell pathology, 2026 College Student Innovation Training Program (2026.05 – present).
- Conference reviewer for **2026 IEEE-RAS 25th International Conference on Humanoid Robots (Humanoids 2026)**, a top conference in humanoid robotics.
- Conference reviewer for **IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2026**, a top robotics conference.
- Head of Organization Department, Future Technology School, SZTU (2024.12 – present).
- Class Monitor & Youth League Secretary (2024.09 – 2025.10).
- Deputy Team Leader, 15th "Challenge Cup" Student Entrepreneurship Competition (2025).

Skills

Mechanism Design:	Parallel-jaw coupled grippers, linear linkages, parallelogram linkages, underactuated & tendon-driven hands; SolidWorks (modeling / assembly / kinematic simulation); Bambu Lab 3D printing; rapid prototyping and grasping experiment design.
Programming:	Python; basic experience with deep learning frameworks (point cloud / quantum circuit simulation).
Academic Writing:	IEEE journal & conference formatting; experienced with the full submission pipeline.
Languages:	Chinese (native); English (CET-4: 580; IELTS scheduled Dec 2026).